

Sruthi Keerthana Nuttacki

Tempe, Arizona | sruthikeerthana960@gmail.com | (602) 730-4639

GitHub | LinkedIn | Portfolio

Education

Arizona State University, Tempe, AZ

Jan 2024 - Dec 2025

M.S., Data Science, Analytics, and Engineering — Electrical Engineering Concentration

Relevant Coursework: Data Mining, Computing for Data-Driven Optimization, Statistics for Data Analysis, Advanced Database Management Systems, Data Visualization, Mathematical Foundations of Machine Learning

Geethanjali College of Engineering and Technology, Hyderabad, India

July 2018 - Sep 2022

B.Tech., Electrical and Electronics Engineering

Technical Skills

Languages: Python, SQL, R, Bash

ML/AI Frameworks: TensorFlow, PyTorch, Scikit-learn, XGBoost, Hugging Face, LangChain, spaCy, NLTK

Data Engineering: PySpark, Pandas, NumPy, Tableau, Power BI, Streamlit

Tools & Infrastructure: AWS, Azure, Docker, Kubernetes, Databricks, GitHub, BeautifulSoup, Selenium

Specializations: Agentic AI, Multi-Agent Systems, LLMs, Generative AI, NLP, Deep Learning, MLOps

Certifications: Computational Data Science, UpGrad INSOFE & Case Western Reserve University, 2022 | Machine Learning I, Columbia University, 2025

Professional Experience

Graduate Research Assistant, Arizona State University, Tempe, AZ

Feb 2026 – Present

- Designing AI-driven Technology Readiness Level assessment framework for MARVEL nuclear microreactor using multi-agent architecture with Python, LangChain, and LLMs to automate synthesis of 100+ technical documents from DOE publications and conference proceedings.
- Building document ingestion pipelines using BeautifulSoup, Selenium, and PyPDF2; applying NLP entity extraction and semantic similarity analysis to identify milestones across six microreactor subsystems.

Machine Learning Engineer, INSOFE, UpGrad, Hyderabad, India

Oct 2021 – Apr 2023

Progression: Trainee to Intern to Full-Time Engineer

- Built production no-code NLP platform using Streamlit for text classification, NER, and sentiment analysis on 10,000+ documents via modular TensorFlow/Keras pipelines with spaCy and NLTK preprocessing.
- Led customer segmentation for Caddo Holdings in Dallas, TX analyzing 50,000+ records using K-means, RFM, Random Forest, and XGBoost, achieving 20% improvement in marketing ROI.
- Developed SQL-backed Tableau dashboards visualizing 15+ KPIs, automating workflows and reducing analysis time by 40 hours per month; supported ML deployments using PySpark with Kubernetes and AWS.

Academic Projects

Agentic AI for Medical Affairs Intelligence in Breast Cancer — Graduate Capstone

- Architected multi-agent AI system processing 129 articles from 50+ sources including FDA, ASCO, ESMO, and pharmaceutical newsrooms to generate 432 HCP-targeted insights, achieving 90% precision using LLMs, spaCy NER, and semantic validation.
- Designed 5-agent pipeline with Scraper, Entity Extractor, Categorizer achieving 75% accuracy across 8 medical categories, Question Generator with grounding score 0.628, and Publisher Agent; implemented three-tier quality filtering enabling Medical Affairs teams to track healthcare professional perceptions in real-time.

Hybrid CNN-LSTM for Predictive Maintenance

- Developed a hybrid CNN-LSTM model with 90% precision to accurately predict machine failures, significantly reducing downtime by 30%, supported by advanced noise reduction and multi-scale feature extraction techniques.
- Automated anomaly detection with t-SNE visualization, accelerating predictive maintenance via cloud-based monitoring.

GPU-Accelerated Vehicle Routing Optimization

- Designed a GPU-accelerated Julia optimization framework using Column Generation, effectively reducing routing computation time by 15% while incorporating complex logistics constraints and dynamic operational variables.
- Validated routing improvements with stochastic benchmarking and metaheuristic analysis for real-time fleet dispatching.

Additional Experience

Admissions Supervisor, 20 hrs/week, Arizona State University, Tempe, AZ

June 2024 – Dec 2025

- Supervised and mentored a high-performing team in administrative operations, improving data accuracy by 20% through data-driven workflow automation and compliance management, demonstrating strong project and process improvement capabilities.

Assistant Recorder, Arizona Digital Health Symposium, ASU, Phoenix, AZ

May 2025

- Coordinated event logistics and meticulously documented key insights during multiple breakout sessions; actively supported interdisciplinary collaboration and contributed to comprehensive post-event reporting for strategic planning.